Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Explainable Audio AI for Human–Machine Interaction			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services			
	Medium Category	Small Category	Medium Category	Small Category
	Human Resource Competency Development Services	Human–System Interaction Technologies		
Concept of Proposed Technology	Current industrial acoustic monitoring systems increasingly rely on audio-based anomaly detection to identify abnormal machine operating conditions [1,2], but they primarily function as black-box detectors, providing limited insight into why specific sounds are flagged as abnormal. As a result, operators often struggle to assess the reliability of alerts, leading to mistrust, unnecessary interventions, or missed faults. Moreover, fully automated systems may fail under novel or complex acoustic conditions, while purely human-based monitoring is prone to fatigue and subjective error. To address these limitations, this technology proposes an explainable audio-based AI framework for human–machine interaction, where machine listening is tightly integrated with human decision-making. The system continuously monitors production-line sounds, detects relevant acoustic events, and generates human-understandable explanations that support operators in evaluating, validating, and responding to detected anomalies. By combining audio intelligence with explainable			

Rationale for Joint R&D Project	AI [3-5] and human-in-the-loop interaction, the proposed approach enables safe automation where appropriate while preserving human oversight for complex or safety-critical decisions.
	Audio-based AI enables cost-efficient and non-intrusive monitoring of industrial production lines, but practical deployment remains challenging due to acoustic variability, limited labeled data, and the need to integrate AI output with human decision-making. These challenges are especially relevant for SMEs operating diverse and evolving production environments. The proposed Explainable Audio AI for Human–Machine Interaction requires joint R&D to combine advanced research on interpretable acoustic modeling and explainable decision support with real-world industrial validation. A single organization cannot independently address both algorithmic development and field-level deployment requirements. The research institute will lead the development of audio AI algorithms, explainability methods, and HMI frameworks, while SME partners will provide production environments, domain expertise, and pilot deployment for system evaluation.
Key Words	1. Explainable Audio AI 2. Human–Machine Interaction (HMI) 3. Acoustic Anomaly Detection 4. Human-in-the-Loop Decision Support 5. Industrial Sound Monitoring

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 7
R&D Objectives	1.) Development of Explainable Audio-Based Anomaly Detection Models: Develop audio AI models capable of detecting abnormal production-line sounds while providing interpretable explanations based on time–frequency representations and acoustic features, enabling transparent understanding of detected events. 2.) Design of Human-in-the-Loop Decision Support Mechanisms: Develop decision-support logic that integrates explainable AI outputs with human interaction, allowing operators to review, validate, and respond to detected acoustic events according to operational risk and complexity. 3.) Implementation of an Industrial Human–Machine Interaction Interface: Develop an HMI framework that presents acoustic analysis results and explanations through intuitive visual and audio feedback, supporting efficient operator interaction under real production conditions. 4.) Validation in Real Production Environments: Validate the proposed technology through pilot deployment on SME production lines to evaluate detection performance, explanation effectiveness, and human decision-making support.			
R&D Target Specifications	The target performance specifications of this project are defined based on comparative evaluation between a baseline audio AI system without explainable interaction and the proposed explainable HMI-enabled system. 1.) Improve operator understanding of AI alerts, achieving at least a 25% increase in correct interpretation rate of detected acoustic events compared to non-explainable audio monitoring systems. 2.) Reduce operator decision time by 20% or more when responding to acoustic alerts, measured as the average time from alert presentation to final human decision. 3.) Reduce incorrect human decisions (false confirmations or unnecessary interventions) by at least 30%, compared to decisions made using non-explainable AI outputs. 4.) Improve human–AI collaboration efficiency by enabling automated handling of at least 50% of predefined routine acoustic events, while ensuring that all novel or high-impact events are explicitly presented to human operators with explainable evidence for final decision-making.			
R&D Plan	**Phase 1: Explainable Audio Modeling and Baseline System Development** In Phase 1, the project focuses on developing core audio-based anomaly detection models and establishing a baseline system for comparative HMI evaluation. The main research objective is to identify acoustic representations and model architectures that support both reliable detection and post-hoc interpretability. The research will investigate time–frequency representations and interpretable acoustic features (e.g., spectral energy distribution, temporal patterns) that can be linked to machine operating states. Baseline anomaly detection models without explainable interaction will be implemented to serve as reference systems for later HMI evaluation. In parallel, initial explainability mechanisms, such as feature attribution and saliency mapping on spectrograms, will be explored to assess feasibility and			

computational cost.

The main milestone of this phase is the development of a baseline audio anomaly detection system and preliminary explainable audio outputs, validated using simulated anomalies and public industrial sound datasets. This phase establishes the technical foundation and evaluation baseline for subsequent HMI-focused research.

Phase 2: Explainable Human–Machine Interaction and Human-in-the-Loop Design

Phase 2 focuses on integrating explainable audio AI into a human–machine interaction framework and studying how explanations support human decision-making. The key research question is how to present acoustic explanations in a form that improves operator understanding, trust, and response efficiency in line with established human–AI interaction principles [6].

This phase will develop multi-level explanation mechanisms, including signal-level visualization, feature-level interpretation, and decision-level summaries, building on recent advances in explainable artificial intelligence for audio signals [7]. Human-in-the-loop decision support logic will be designed to distinguish between routine, low-risk events and complex or high-impact cases requiring human validation. HMI prototypes will be developed to enable interactive exploration of detected events, such as replaying sounds and comparing normal versus abnormal acoustic patterns.

Controlled user studies will be conducted to compare operator performance using the baseline system (without XAI) and the proposed explainable HMI system. Key metrics will include interpretation accuracy, decision time, and error rates.

The milestone of this phase is the demonstration of measurable improvements in human decision-making performance enabled by explainable audio-based interaction.

Phase 3: Adaptive Automation and Field Validation in SME Environments

Building on the explainable HMI framework developed in earlier phases, Phase 3 aims to validate the system in real production environments and evaluate its readiness for deployment. This phase focuses on adaptive automation, where the level of human involvement is adjusted based on event risk and system confidence.

The explainable audio AI system will be deployed on production lines provided by SME partners. Routine acoustic events will be handled through automated or semi-automated workflows, while complex events will trigger explainable alerts requiring human decisions. Operator feedback will be logged to refine system behavior and interaction policies.

Field-level evaluation will be conducted using practical KPIs, including reduction in response time, decrease in incorrect human decisions, explainability coverage, and system stability under changing acoustic conditions.

The final milestone is the validation of a deployable explainable audio AI system that supports effective human–AI collaboration and adaptive automation in real SME manufacturing environments.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute (Fraunhofer IDMT) will lead the applied research and technical development of the proposed Explainable Audio AI for Human–Machine Interaction, with a primary focus on audio-based monitoring, analysis, and system integration for industrial environments. The institute's role is to develop robust and interpretable audio AI technologies that support reliable machine monitoring and operator decision-making. Specifically, Fraunhofer IDMT will: 1.) Develop and optimize audio-based anomaly detection and event recognition algorithms for industrial machine sounds, building on established expertise in machine listening and acoustic signal processing. 2.) Design explainable audio analysis methods based on time–frequency representations and interpretable acoustic features, enabling transparent interpretation of detected events by human operators. 3.) Implement engineering-oriented human–machine interaction mechanisms for presenting acoustic alerts and explanations, including visual and audio-based feedback suitable for industrial use. 4.) Develop decision-support logic that enables practical collaboration between automated audio monitoring and human operator intervention, with a focus on safe and efficient operation rather than full automation. 5.) Conduct systematic technical evaluations and field-oriented validation, including comparative analysis of system performance with and without explainable interaction. In addition, Fraunhofer IDMT will provide research infrastructure, computing resources, and expertise in audio analysis, machine learning, and applied system engineering. The institute will support knowledge transfer through technical documentation, prototype implementations, and contributions to joint intellectual property and standardization-relevant activities.
Expected R&R of Korean SME Partner	The Korean SME partner will lead the industrial application, field validation, and commercialization-oriented development of the proposed Explainable Audio AI for Human–Machine Interaction technology. While the research institute focuses on audio AI and explainability methods, the SME partner will ensure that the developed system is practical, deployable, and aligned with real manufacturing requirements. Specifically, the Korean SME partner will: 1.) Provide access to production-line environments, machinery, and representative acoustic data for system development, testing, and validation. 2.) Support system integration and field deployment, including sensor installation, operational configuration, and day-to-day system operation. 3.) Define practical performance requirements and operational KPIs, such as acceptable false alarm rates, response time constraints, and automation thresholds. 4.) Participate in the evaluation of human–machine interaction by involving operators in pilot testing, feedback collection, and comparative assessment of system usability with and without explainable interaction. 5.) Contribute to productization and commercialization planning, including integration into existing industrial solutions and identification of target markets and customers. Through these roles, the Korean SME partner will ensure that the proposed technology is validated under real operating conditions and positioned for sustainable adoption and commercialization in SME manufacturing environments.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to deliver both technological advancements in audio-based AI systems and practical value for industrial SMEs. From a technical perspective, the project will establish applied methodologies for integrating explainable audio analysis with human–machine interaction, enabling transparent and reliable interpretation of production-line sounds. The resulting system will demonstrate how audio-based anomaly detection can be combined with explainable decision support and adaptive automation in real industrial environments. From a commercialization perspective, the developed technology will support SMEs and solution providers in deploying acoustic monitoring systems with higher operational trust and reduced risk. By enabling operators to understand and validate AI-generated alerts, the proposed approach reduces false interventions and accelerates decision-making, facilitating faster adoption and integration into existing industrial solutions. The project outcomes are expected to support productization of explainable audio monitoring modules that can be tailored to different manufacturing contexts. At the industrial and economic level, the joint R&D project will contribute to improved production reliability, reduced downtime, and enhanced operational safety in SME manufacturing. Through international collaboration, the project will strengthen technology transfer, support the global competitiveness of Korean SMEs, and promote the adoption of trustworthy, human-centered AI technologies in industrial applications.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization